Project: Worked examples
Title: Pipe expansion, loops and anchor forces (screening)
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1. Scope and method

Calculates the thermal growth of a pipe run between anchors, the leg or loop that absorbs it, and the force and moment on the anchors. It is a screening calculation, not a piping stress analysis.

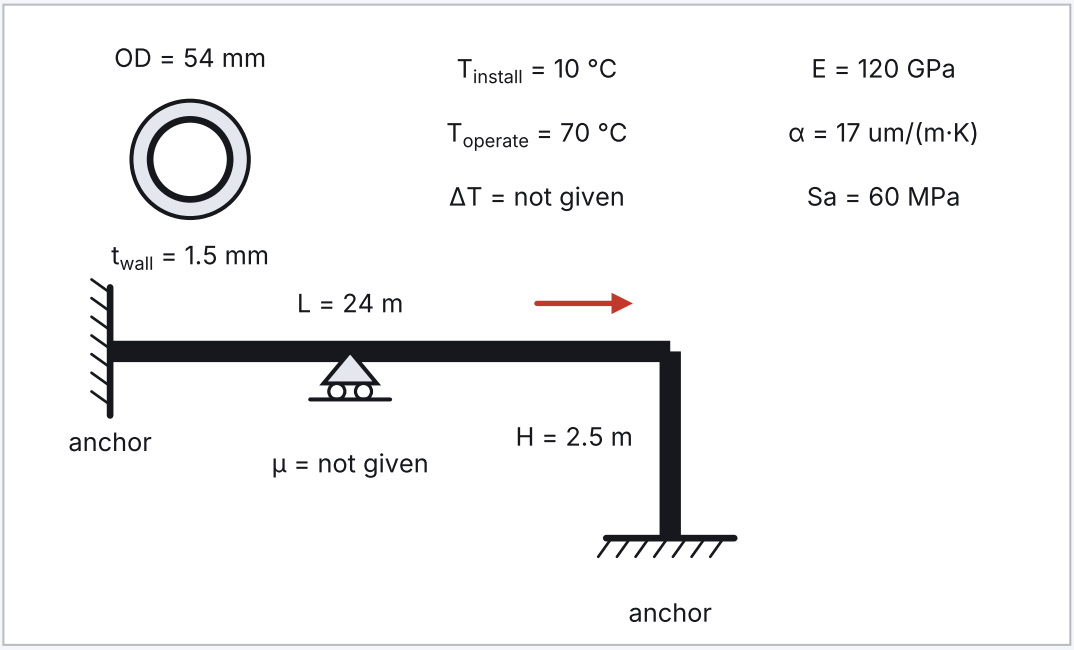


Figure 1 — An offset leg (L-bend): the run L grows from its anchor and bends the leg H at its end; the pipe section beside it. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Pipe section	$d_i = OD - 2 \cdot t$; $A_m = \pi/4 \cdot (OD^2 - d_i^2)$; $I = \pi/64 \cdot (OD^4 - d_i^4)$; $Z = I/(OD/2)$
Temperature change and growth	$\Delta T = T_{operate} - T_{install}$, or as entered; L_{tot} = L, or L + L2 for a Z-bend; $\Delta L = \alpha \cdot L_{tot} \cdot \lvert\Delta T\rvert$, its sign giving the direction
Fully restrained pipe	$\sigma_r = E \cdot \alpha \cdot \lvert\Delta T\rvert$ (E at the installation temperature); $F_r = E_r \cdot \alpha \cdot \lvert\Delta T\rvert \cdot A_m$ ($E_r = E_{hot}$, else E)
Guided cantilever, required leg	Δ = ΔL for an offset leg or Z-bend and ΔL/2 for a U-loop; $L_{req} = \sqrt{(3 \cdot E \cdot OD \cdot \Delta/Sa)}$; PASS when $L_{req} \le$ the available leg H
Force and moment of the available leg	$F_{leg} = 12 \cdot E_r \cdot I \cdot \Delta/H^3$ (along the run); $M_{leg} = 6 \cdot E_r \cdot I \cdot \Delta/H^2$ (= $F_{leg} \cdot H/2$); $\sigma_{leg} = 3 \cdot E \cdot OD \cdot \Delta/H^2 = Sa \cdot (L_{req}/H)^2$
Runs of an offset leg or Z-bend	the runs bend for the leg's growth $\Delta_{leg} = \alpha \cdot \lvert\Delta T\rvert \cdot H$ under one force $F_{run} = 12 \cdot E_r \cdot I \cdot \Delta_{leg}/(L^3 + L2^3)$ along the leg (L2 = 0 for an L-bend); run i takes $L_i^3/(L^3 + L2^3)$ of Δ_{leg}, with $\sigma_i = 3 \cdot E \cdot OD \cdot \Delta_i/L_i^2$ and $M_i = F_{run} \cdot L_i/2$, and passes when $\sqrt{(\sigma_i/Sa)} \le 1$

WHAT IT CALCULATES	RELATION
Force on the anchors	$\sqrt{(F_{leg}^2 + F_{run}^2)}$ from the two perpendicular components; the moment reported is the largest root moment of the legs and runs
Screening and bellows	$OD \cdot Y/(L_{dev} - U)^2 \le K1$ with $Y = \alpha \cdot \lvert\Delta T\rvert \cdot U$ (offset leg or Z-bend $L_{dev} = L_{tot} + H$, $U = \sqrt{(L_{tot}^2 + H^2)}$; U-loop $L_{dev} = L + 2 \cdot H$, $U = L$); axial bellows anchor force = $P \cdot A_{eff} + k \cdot \Delta L$, with no leg check and no screening

Basis: Process piping — flexibility analysis, exemption from formal analysis[1](cl. 319.4.1 (screening relation; number to be checked against the owner's copy)); Process piping — displacement stress range and reactions[1](cl. 319.3.2 and 319.5 (reference and hot moduli; numbers to be checked against the owner's copy)); Metallic industrial piping — flexibility analysis[2](cl. 12 (number to be checked against the owner's copy)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
layout	What absorbs the movement	offset leg (L- or Z-bend: the leg takes the whole movement)		drawing: DRG-M-212 rev A	any
$temp_{mode}$	Temperature change from	install and operating temperatures		assumption: Install and operating temperatures are both known.	any
OD	Pipe outside diameter	54	mm	drawing: DRG-M-212 rev A	10 … 1500 mm
t_{wall}	Pipe wall thickness	1.5	mm	drawing: DRG-M-212 rev A	0.5 … 60 mm
L	Anchor spacing: the straight length that grows (an offset leg or Z-bend: the first run, which also bends for the leg's growth)	24	m	drawing: DRG-M-212 rev A	1 … 500 m
H	Available leg length (each leg of a U-loop; an offset leg: the leg off the run, which may be the longer of the two); optional for an axial bellows, which has no leg; no default	2.5	m	drawing: DRG-M-212 rev A	0.5 … 50 m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
E	Young's modulus of the pipe material at the installation (reference) temperature, for the stress check, in GPa (steel 210, copper 120, PE and PP about 1; not MPa)	120	GPa	assumption: W3 seed 'Copper (generic)', indicative; replace with the project's material record.	0.05 ... 500 GPa
α	Mean coefficient of thermal expansion between the install and operating temperatures	17	um/(m·K)	assumption: W3 seed 'Copper (generic)' (17e-6 1/K), indicative; replace with the project's material record.	1 ... 250 um/(m·K)
$T_{install}$	Temperature at installation (when the run is anchored)	10	°C	assumption: Mean site temperature when the riser is anchored.	-50 ... 50 °C
$T_{operate}$	Operating temperature (hot or cold)	70	°C	spec: SPEC-M-04 §3: heating flow temperature	-50 ... 500 °C
Sa	Allowable displacement-stress range for the legs and runs (your piping code and its formula; optional for an axial bellows; no default)	60	MPa	assumption: Invented illustrative allowable range; take Sa from the project's piping code and its formula.	5 ... 500 MPa
K_{scr}	Screening limit K1 for OD·Y/(L − U)², in mm²/m² (or in²/ft², ...) from your piping code (optional; blank = screening not evaluated, so the sheet does not pass; no default)	100	mm²/m²	assumption: Invented illustrative screening limit; take K1 from the project's piping code.	1 ... 1000 mm²/m²

3. Assumptions

A1. The result is a screen, not a piping stress analysis. (Stated scope.)
A2. The result tells whether a simple leg or loop is plausible and what it puts on its anchors. (Stated scope.)
A3. It does not replace a flexibility analysis of the real routing, including elbow flexibility, intensification, torsion, branches, supports and loads. (Stated scope.)
A4. A guided cantilever is straight, uniform, fixed against rotation at both ends and loaded only by the movement it absorbs. (Beam theory.)
A5. Its stiffness is 12·E·I/H³ and its end moment is F·H/2. (Beam theory.)
A6. Elbows are taken as rigid and the leg as slender. (Beam theory.)
A7. The beam model is not valid for legs shorter than about ten diameters. (Beam theory.)
A8. No stress-intensification factor and no elbow flexibility are applied. (Stated limit.)

A9. The stress at the elbow root of a real bend is higher than the beam stress M/Z. (Stated limit.)
A10. A leg that passes here can still exceed Sa at the elbow, which is why the result is a screen. (Stated limit.)
A11. A U-loop is two identical legs in series, each a guided cantilever taking half the growth. (Loop model.)
A12. The cross leg of a U-loop is a rigid guide. (Loop model.)
A13. The force along the line is the same in both legs of a U-loop. (Loop model.)
A14. The moment M is the moment at the leg roots, the corners of the loop. (Loop model.)
A15. It reaches an anchor only where the leg ends at the anchor, as in an offset leg. (Loop model.)
A16. The growth of the loop's own legs, which the cross leg has to absorb, is not checked. (Loop model.)
A17. The run of an offset leg bends for the leg's growth as a second guided cantilever, checked against Sa. (Offset-leg and Z-bend model.)
A18. In a Z-bend the two runs are in series under one force. (Offset-leg and Z-bend model.)
A19. Each run takes its share of the leg's growth in proportion to L_i^3 and is checked as its own cantilever. (Offset-leg and Z-bend model.)
A20. The leg of a Z-bend takes the growth of both runs. (Offset-leg and Z-bend model.)
A21. The force on the anchors is the resultant of the two perpendicular forces along the run and along the leg. (Offset-leg and Z-bend model.)
A22. $\text{anchor}_{\text{moment}}$ is the largest root moment of the legs and runs, M = F·L/2 with each end fixed against rotation. (Stated limit.)
A23. It is not the moment on a frame anchor, which depends on the whole frame. (Stated limit.)
A24. A formal analysis is needed before an anchor is sized for moment. (Stated limit.)
A25. α is the mean coefficient of expansion between the install and operating temperatures. (Material library.)
A26. E is the modulus at the installation temperature, used for the stress check as the piping codes do. (The ASME process piping code, cl. 319.3.2 and 319.5 (numbers to be checked against the owner's copy).)
A27. E_{hot}, when given, is used only for the reactions on the anchors. (Code convention to be checked.)
A28. The material properties are taken from a library material applied at the right temperature. (Material library.)
A29. Sa and K1 are the owner's records from the owner's own copy of the piping code or the design basis. (No table from a standard is shipped.)
A30. No allowable stress, K1 or code constant ships. (No table from a standard is shipped.)
A31. The sheet applies the relation to the numbers entered, whatever code or load case they come from. (No table from a standard is shipped.)
A32. No weight, spring-hanger or cold-spring effect is included in the leg force. (Stated limit.)
A33. The anchor force adds support friction μ·w·L only when μ and the weight per metre are entered. (Stated limit; a warning says so when they are not.)
A34. The bellows anchor force is the pressure thrust $P \cdot A_{\text{eff}}$ plus the spring force k·ΔL at the governing pressure. (Stated limit.)
A35. The governing pressure is the design or test pressure, whichever governs, because compression and extension ratings can differ. (Stated limit.)
A36. Guide friction and any other bellows load come from the manufacturer's data. (Stated limit.)
A37. $\text{temp}_{\text{mode}}$ (Temperature change from) is taken as install and operating temperatures. (Install and operating temperatures are both known.)
A38. E (Young's modulus of the pipe material at the installation (reference) temperature, for the stress check, in GPa (steel 210, copper 120, PE and PP about 1; not MPa)) is taken as 120 GPa. (W3 seed 'Copper (generic)', indicative; replace with the project's material record.)
A39. α (Mean coefficient of thermal expansion between the install and operating temperatures) is taken as 17 um/(m·K). (W3 seed 'Copper (generic)' (17e-6 1/K), indicative; replace with the project's material record.)
A40. T_{install} (Temperature at installation (when the run is anchored)) is taken as 10 °C. (Mean site temperature when the riser is anchored.)
A41. Sa (Allowable displacement-stress range for the legs and runs (your piping code and its formula;

optional for an axial bellows; no default)) is taken as 60 MPa. (Invented illustrative allowable range; take Sa from the project's piping code and its formula.)
A42. K_{scr} (Screening limit K1 for OD·Y/(L − U)², in mm²/m² (or in²/ft², …) from your piping code (optional; blank = screening not evaluated, so the sheet does not pass; no default)) is taken as 100 mm²/m². (Invented illustrative screening limit; take K1 from the project's piping code.)
A43. Layout: the offset leg of 2.5 m takes the whole movement of 24.48 mm as a guided cantilever fixed against rotation at both ends, and the run of 24 m bends for the leg's growth of 2.55 mm. Screening only — not a piping stress analysis. (Guided-cantilever model.)
A44. The available leg (2.5 m) is shorter than the 2.8163 m required: the bending stress at its root is 76.143 MPa against Sa = 60 MPa. Lengthen the leg or loop, add a loop or bellows, or analyse the line formally. (Result of the check.)

4. Working

[9] $dT = T_{operate} - T_{install}$
$= 70\ degC - 10\ degC$
$= 60\ K$
Temperature change, operating − install
[11] $E_r = E$
$= 120\ GPa$
$= 120\ GPa$
Modulus for the reactions (E, no E_{hot} given)
[12] $D_i = OD - 2 * t_{wall}$
$= 54\ mm - 2 \times 1.5\ mm$
$= 51\ mm$
Inside diameter
[13] $A_m = pi / 4 * (OD^2 - D_i^2)$
$= \pi / 4 \times ((54\ mm)^2 - (51\ mm)^2)$
$= 247.4\ mm^2$
Metal cross-section
[14] $I_x = pi / 64 * (OD^4 - D_i^4)$
$= \pi / 64 \times ((54\ mm)^4 - (51\ mm)^4)$
$= 8.5307\ cm^4$
Second moment of area of the pipe section
[15] $Z_x = I_x / (OD / 2)$
$= 8.5307\ cm^4 / (54\ mm / 2)$
$= 3.1595\ cm^3$
Section modulus
[16] $L_{tot} = L$
$= 24\ m$
$= 24\ m$
Length that grows between the anchors
[17] $dL = \alpha * L_{tot} * abs(dT)$
$= 17\ um/(m \cdot K) \times 24\ m \times abs(60\ K)$
$= 24.48\ mm$
Thermal growth between the anchors, as a magnitude
[18] $dL_{signed} = \alpha * L_{tot} * dT$
$= 17\ um/(m \cdot K) \times 24\ m \times 60\ K$
$= 24.48\ mm$
Thermal movement with its sign: positive lengthens, negative shortens
[19] $\sigma_{res} = E * \alpha * abs(dT)$
$= 120\ GPa \times 17\ um/(m \cdot K) \times abs(60\ K)$
$= 122.4\ MPa$
Stress in the fully restrained pipe (E at the installation temperature)

[20] $F_{res} = E_r * \alpha * abs(dT) * A_m$
= 120 GPa × 17 um/(m·K) × abs(60 K) × 247.4 mm²
= 30.282 kN
Force on the anchors of the fully restrained pipe
[21] $F_{friction}$ = support friction not entered
= μ and w_{pipe} blank
= 0 kN
Friction of the sliding supports (not entered: none in the anchor force)
[24] $u_{res} = \sigma_{res} / Sa$
= 122.4 MPa / 60 MPa
= 2.04
Fully restrained stress as a fraction of Sa (above 1: the movement has to be absorbed)
[25] Δ = dL
= 24.48 mm
= 24.48 mm
Movement the offset leg absorbs (all of it)
[26] $L_{req} = sqrt(3 * E * OD * \Delta / Sa)$
= sqrt(3 × 120 GPa × 54 mm × 24.48 mm / 60 MPa)
= 2.8163 m
Leg length at which the guided-cantilever bending stress reaches Sa
[27] $u_{leg} = L_{req} / H$
= 2.8163 m / 2.5 m
= 1.1265
Required leg over available leg
[28] $F_{leg} = 12 * E_r * I_x * \Delta / H^3$
= 12 × 120 GPa × 8.5307 cm⁴ × 24.48 mm / (2.5 m)^3
= 0.19246 kN
Force of the available leg on its anchors or guides
[29] $M_{leg} = 6 * E_r * I_x * \Delta / H^2$
= 6 × 120 GPa × 8.5307 cm⁴ × 24.48 mm / (2.5 m)^2
= 0.24057 kN·m
Moment at the root of the available leg
[30] $\sigma_{leg} = 3 * E * OD * \Delta / H^2$
= 3 × 120 GPa × 54 mm × 24.48 mm / (2.5 m)^2
= 76.143 MPa
Bending stress at the root of the available leg (E at the installation temperature)
[31] $u_{sigma} = \sigma_{leg} / Sa$
= 76.143 MPa / 60 MPa
= 1.269
Bending stress over Sa (= u_{leg}^2)
[32] $\Delta_{run} = \alpha * abs(dT) * H$
= 17 um/(m·K) × abs(60 K) × 2.5 m
= 2.55 mm
Growth of the leg, which the run or runs have to absorb
[33] $S_{run} = L^3$
= (24 m)^3
= 13824 m³
Cube of the run (its flexibility)
[34] $F_{run} = 12 * E_r * I_x * \Delta_{run} / S_{run}$
= 12 × 120 GPa × 8.5307 cm⁴ × 2.55 mm / 13824 m³
= 0.00002266 kN
Force of the run or runs, along the leg

[35] $\sigma_{run} = 3 * E * OD * \Delta_{run} * L / S_{run}$
= 3 × 120 GPa × 54 mm × 2.55 mm × 24 m / 13824 m³
= 0.086062 MPa
Bending stress at the root of the run
[36] $u_{run} = sqrt(\sigma_{run} / Sa)$
= sqrt(0.086062 MPa / 60 MPa)
= 0.037873
Required run over available run (= √(σ/Sa))
[37] $L_{req,run} = L * u_{run}$
= 24 m × 0.037873
= 0.90896 m
Cantilever length for this run's share of the leg's growth
[38] $M_{run} = F_{run} * L / 2$
= 0.00002266 kN × 24 m / 2
= 0.00027192 kN·m
Moment at the root of the run
[39] $F_{along,run} = F_{leg} + F_{friction}$
= 0.19246 kN + 0 kN
= 0.19246 kN
Component of the anchor force along the run: the leg's force plus the support friction
[40] $F_{along,leg} = F_{run}$
= 0.00002266 kN
= 0.00002266 kN
Component of the anchor force along the leg: the runs' force
[41] $anchor_{force} = sqrt(F_{along,run}^2 + F_{run}^2)$
= sqrt((0.19246 kN)^2 + (0.00002266 kN)^2)
= 0.19246 kN
Force on the anchors: resultant of the two perpendicular components (support friction only if μ and w_{pipe} are entered)
[42] $anchor_{moment} = max(M_{leg}, M_{run})$
= max(0.24057 kN·m, 0.00027192 kN·m)
= 0.24057 kN·m
Largest root moment of the leg and the run or runs (guided-cantilever model)
[43] $L_{dev} = L_{tot} + H$
= 24 m + 2.5 m
= 26.5 m
Developed length between the anchors, run or runs plus leg
[44] $U_a = sqrt(L_{tot}^2 + H^2)$
= sqrt((24 m)^2 + (2.5 m)^2)
= 24.13 m
Straight distance between the anchors
[45] $Y_{scr} = \alpha * abs(dT) * U_a$
= 17 um/(m·K) × abs(60 K) × 24.13 m
= 24.612 mm
Resultant displacement Y: the free growth between the anchors
[46] $scr = OD * Y_{scr} / (L_{dev} - U_a)^2$
= 54 mm × 24.612 mm / (26.5 m − 24.13 m)^2
= 236.59 mm²/m²
Screening ratio OD·Y/(L − U)²
[48] $u_{scr} = scr / K_{scr}$
= 236.59 mm²/m² / 100 mm²/m²
= 2.3659
Screening ratio over the K1 record

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[49] pass = K1 given and every check passes
= 3 check(s)
= 0
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1 when the leg, the run (offset leg) and the screening checks all pass; 0 when one fails or when no K1 was entered

5. Results

SYMBOL	VALUE	DESCRIPTION
$anchor_{force}$	0.1925 kN	Force on the anchors (resultant)
$anchor_{moment}$	0.2406 kN·m	Largest root moment (guided-cantilever model)
dL	24.48 mm	Thermal movement between the anchors (magnitude)
D_i	51 mm	Inside diameter
A_m	247.4 mm²	Metal cross-section
I_x	8.531 cm⁴	Second moment of area
Z_x	3.160 cm³	Section modulus
dT	60 K	Temperature change
L_{tot}	24 m	Length that grows
dL_{signed}	24.48 mm	Thermal movement, signed
σ_{res}	122.4 MPa	Fully restrained stress
F_{res}	30.28 kN	Fully restrained force
direction	"growth: the pipe lengthens; fully restrained it presses on its anchors (compression)"	Direction of the movement
$F_{friction}$	0 kN	Support friction on the run
Δ	24.48 mm	Movement absorbed by each leg
L_{req}	2.817 m	Required leg length
F_{leg}	0.1925 kN	Force of the leg
M_{leg}	0.2406 kN·m	Moment at the leg root
σ_{leg}	76.14 MPa	Bending stress at the leg root
Δ_{run}	2.550 mm	Leg growth absorbed by the run or runs
S_{run}	13824 m³	Flexibility of the runs
F_{run}	0.00002266 kN	Force of the runs
σ_{run}	0.08606 MPa	Bending stress at the first run root
$L_{req,run}$	0.9090 m	Required length of the first run
M_{run}	0.0002719 kN·m	Moment at the first run root
$F_{along,run}$	0.1925 kN	Anchor force component along the run
$F_{along,leg}$	0.00002266 kN	Anchor force component along the leg
L_{dev}	26.50 m	Developed length

SYMBOL	VALUE	DESCRIPTION
U_a	24.13 m	Anchor distance
Y_{scr}	24.61 mm	Resultant displacement Y
scr	236.6 mm²/m²	Screening ratio OD·Y/(L − U)² (governing)
screen	"needs formal stress analysis"	Screening result
pass	0	All checks pass
u_{res}	2.04	Fully restrained stress over Sa
u_{leg}	1.13	Required leg over available leg
u_{sigma}	1.27	Bending stress over Sa
u_{run}	0.04	First run: required over available
u_{scr}	2.37	Screening ratio over K1

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Screening ratio against the limit K1 (not met: the line needs a formal stress analysis)	236.6 mm²/m²	100 mm²/m²	2.37	[1](cl. 319.4.1 (screening form; number to be checked against the owner's copy))
FAIL	Available leg against the required leg	2.817 m	2.500 m	1.13	
PASS	Run against the length it needs to absorb the leg's growth	0.9090 m	24 m	0.04	

7. Applicability

WARNING records: tabulated values not taken from a reference record are E, α (Material properties); OD, t_{wall} (Pipe section (outside diameter and wall)); Sa (Allowable displacement-stress range); K_{scr} (Screening limit K1). Enter each as one so its source, edition and licence travel with it. [record]
WARNING μ: the anchor force holds no support friction. Enter μ and the weight per metre, with the length on slides, to include it. [friction]

8. Verification

Template bs.pipe-expansion@1.5.2 Content hash 037fa1e46b3af76f Engine calc-engine/1.0.0
Known-answer cases passed 6/6 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. ASME B31.3-2022.
2. BS EN 13480-3:2017.